

Dr. Vahram Martirosyan

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Abu Dhabi, UAE

Chief Scientist · Assistant Professor · Explainable AI · HPC & Compiler Engineering · Academic Leader

Teaching Philosophy & Profile

Ph.D. in Computational Mathematics with **15+ years of classroom teaching** across universities in Armenia and the UAE, combined with 20+ years of professional software engineering and active research. I believe rigorous computer-science education must connect theory to practice: my courses consistently move students from first principles to hands-on implementation using real systems.

Current areas of instruction span **programming (C, C++, Python)**, **system architecture, data structures & algorithms**, **operating systems**, **IoT & security**, and **statistics**. Active researcher in intrinsically **explainable AI** via polynomial regression models, applied to medicine (ECG and clinical diagnosis), materials science, precision agriculture, and ocean/climate science, and in **HPC & compiler engineering**: LLVM-based automatic parallelisation (HPC++) and binary lifting of legacy code (LEAP). 25+ publications, including a published IEEE-indexed conference paper, several manuscripts under review, and a pending patent, with a track record of integrating research directly into the classroom. Currently **Chief Scientist** at [ExplainableAI.app](#) and **Assistant Professor and Course Team Leader** (multi-campus) at Higher Colleges of Technology, Abu Dhabi.

Education

Ph.D., Computational Mathematics 2006 – 2009
Yerevan State University

Dissertation: “Construction of an Algorithm for the Pseudoinversion of Bidiagonal Matrices and an Investigation of its Parallelization”

M.Sc., Informatics and Applied Mathematics 2004 – 2006
Yerevan State University

B.Sc., Informatics and Applied Mathematics 2000 – 2004
Yerevan State University

Courses Taught

- **Programming:** Fundamentals of Programming, C Programming, C++ Programming, High-performance computing in C++, .NET/C# Programming, Intro to Computer Science.
- **Systems & Architecture:** System Architecture & Integration, Computer Architecture I, Operating Systems, Linux/POSIX Programming, Win32 API Programming, Linux Device Driver Development.
- **Data & Mathematics:** Data Structures & Algorithms, Discrete Mathematics, Applied Discrete Mathematics, Database Management Systems, Statistics & Probability.
- **Emerging Topics:** IoT & Security, Computational Thinking, GUI Development (GTK+), System Programming.

Teaching Experience

Course Team Leader

Jan 2025 – Present

Higher Colleges of Technology, Abu Dhabi, UAE

- Managing and guiding a team of lecturers across multiple HCT campuses for the System Architecture & Integration course.
- Developing and maintaining course materials, assessments, and standardised delivery frameworks.

Assistant Professor, Computer Information Science

Jan 2024 – Present

Higher Colleges of Technology, Abu Dhabi, UAE

- Courses: Fundamentals of Programming (Java), System Architecture & Integration, IoT & Security, Data Structures & Algorithms, Statistics & Probability.

Senior Lecturer (Contract)

Feb 2021 – Dec 2023

Yerevan State University, Faculty of Informatics & Applied Mathematics

- Courses: Operating Systems. Supervised undergraduate research projects.

Faculty Lecturer (Contract)

Jan 2023 – Dec 2023

Université française en Arménie (UFAR), Yerevan

- Courses: Computer Architecture I, C Programming.

Visiting Lecturer (Contract)

Aug 2022 – Jan 2023

American University of Armenia, Yerevan

- Course: Introduction to Computer Science.

Informatics Teacher (Contract)

Sep 2023 – Dec 2023

PhysMath School After A. Shahinyan, Yerevan, Armenia

- Taught Informatics at a specialist physics-mathematics secondary school.

Founder & Lecturer

Jan 2019 – Jun 2023

MutEx School, Yerevan

- Founded and ran an independent programming school; designed curricula and delivered public lectures.

Lecturer

Sep 2013 – Dec 2019

Armsoft CS Learning Center, Yerevan

- Course: Operating Systems.

Lecturer

Feb 2015 – Jun 2015

Armenian Code Academy, Yerevan

- Course: .NET / C# Programming.

Senior Lecturer

Feb 2007 – 2014

Russian-Armenian (Slavonic) University, Yerevan

- Courses: Linux/POSIX Programming, Win32 API, Linux Device Driver Development, GUI Software Development (GTK+), Operating Systems, System Programming, Database Management Systems.

Technical Skills

- **Explainable AI (PRMs):** Polynomial Regression Models, feature importance, interpretable classification for clinical, materials science, and agricultural data; Statistics & Probability.
- **Machine Learning:** Supervised learning, regression modelling, interpretable model design, Python, PyTorch.

- **HPC & Parallelism:** LLVM/Clang (compiler pass development), OpenCL, MPI, POSIX Threads, auto-vectorisation, heterogeneous CPU-GPU execution, Xilinx ACAP.
- **Systems Programming:** C/C++ (expert), Linux kernel modules, device drivers (Ext4, XFS, JFS, BtrFS), WIN32 API, GTK+, IoT/embedded systems.
- **Web & Databases:** PHP/MySQL, ASP.NET/C#/MS SQL Server, SQL.
- **Tooling:** Git, Linux/Unix, L^AT_EX.
- **Languages:** Armenian (native), Russian (fluent), English (fluent).

Publications

Published

- **Martirosyan, V.** “Accuracy and Interpretability from a Single Solver: Polynomial Regression up to Degree 5 for Concrete Compressive Strength Prediction.” *ICACTE 2026* (IEEE-indexed conference), 2026.
(Degree-5 $R^2 = 0.859$, RMSE 6.25 MPa; 88% term reduction with 74% of the retained terms aligning with known concrete physics; single closed-form solve, no gradient descent.)

Under Review / Submitted

- **Martirosyan, V.** “LEAP: Lifting Legacy x86-64 Binaries to Re compilable IR for Automatic Parallelization.” *Submitted to Frontiers of Computer Science*, 2026.
(43-program liftability study: 100% at -00 and 98% at -01/02/03; $3.7\times$ geomean speedup from source IR; uplifted binary $10.3\times$ over -00 and $2.1\times$ over -02; $246\times$ peak GPU offload; 377/377 tests pass.)
- **Martirosyan, V.** “On-Device TinyML for Multi-Variable Ocean Profiling” (eICocean). *Submitted to eICocean 2027*, 2026.
($R^2 = 0.973$ temperature, 0.794 salinity, 0.895 dissolved oxygen; $4.4\mu\text{s}$ desktop inference; 1.09 ms on ESP32 and 1.49 ms on RP2350; $180\mu\text{J}/96\mu\text{J}$ per inference.)
- **Martirosyan, V.** “Interpretable Polynomial Zone Ensembles for Global Ocean Temperature and Salinity Profile Reconstruction.” *Under review* (Q1 journal), 2026.
(WOA 2018, 534,090 samples; pooled $R^2 = 0.9703$ temperature / 0.8376 salinity; sub-millisecond inference, sub-megabyte footprint.)
- **Martirosyan, V.** “Depth-Zone Polynomial Regression for Global Ocean Climatology.” *Under review* (IEEE journal), 2026.
- **Martirosyan, V.** “Efficient Multi-Class ECG Classification Using Polynomial Regression with an Analytical Solver.” *Applied Computational Intelligence and Soft Computing* (major revision), 2026.
($82.75\% \pm 1.03\%$ six-class accuracy, macro-F1 0.758; $12\mu\text{s}$ inference; outperforms CNN, random forest, XGBoost and SVM baselines.)
- **Martirosyan, V.** “Automated Multi-Class ECG Arrhythmia Classification Using an Ensemble of Polynomial Regression Models.” *IEEE ICCA 2026* (submitted), 2026.
(78.32% balanced accuracy, 68.23% macro-F1; sparsification reduces 123.1M terms to 330K — 1.3 MB, a 99.7% reduction — within ≈ 1 pp.)
- **Martirosyan, V.** “Sparse Polynomial Regression for Interpretable Yield-Strength Prediction in High-Entropy Alloys.” *Under review* (IEEE ITT 2026), 2026.

(Degree-3 $R^2 = 0.738$ on $n = 1,014$, statistically tied with random forest and XGBoost and significantly above a neural network (0.618); 8 of 11 retained terms map to known hardening mechanisms; BCC-only $R^2 = 0.841$.)

- **Martirosyan, V.** “Efficient Crop Type Mapping via Degree-2 Polynomial Feature Expansion of Multi-Temporal Optical and SAR Data.” *Submitted to IEEE TGRS*, 2026.
(98.67% test accuracy and macro-F1; 2.3 s training on 16 cores; $17\times$ SIMD speedup.)

Preprints & Prepared Manuscripts

- **Martirosyan, V.** “hpc++: An LLVM-Based Automatic Parallelization System for C++ with GPU Acceleration.” arXiv preprint, Dec 2025.
(7 parallelisation strategies, zero source modification, 24 benchmarks; $16.8\times$ average, $97\times$ peak CPU, $247\times$ peak GPU; 100% numerical correctness.)
- **Martirosyan, V.** “Auditable Kilobyte-Scale Surrogates for the Global Significant-Wave-Height Climatology.” *Prepared for IEEE JSTARS*, 2026.
($R^2 = 0.945$, RMSE 0.262 m; cell-holdout $R^2 = 0.933$; coastal R^2 improves from 0.34 to 0.52 with in-situ buoys.)
- **Martirosyan, V.** “Interpretable Polynomial Zone Ensembles for Global Ocean Temperature and Salinity Profile Reconstruction.” *Prepared for IEEE JSTARS*, 2026.
- **Martirosyan, V.** “Polynomial Regression Models for Interpretable Machine Learning: Theory, Optimization, and Medical Applications.” Feb 2026.
- **Martirosyan, V.** “Stagewise Regularized Polynomial Regression: A Hybrid Optimization Strategy for Medical Diagnosis on Small Datasets.” Feb 2026.
(86.7% accuracy on UCI Cleveland Heart Disease; 93% model compression; ≈ 36 FLOPs/prediction.)
- **Martirosyan, V.** “Heart Disease Diagnosis Using Third-Order Polynomial Regression Models with L2 Regularization.” Feb 2026.
(81.8% accuracy, 10-fold CV; full interpretability via explicit polynomial coefficients.)
- **Martirosyan, V.** “Explainable and Efficient Digit Classification via Polynomial Regression Networks: A Transparent Alternative to Deep Learning.” arXiv preprint, Jan 2026.
(96.85% average MNIST accuracy; 84% parameter reduction at $<1\%$ accuracy loss.)
- **Martirosyan, V.** “Blended Ocean Visualization: Combining Sparse Polynomial Regression with Residual Fields.” Working paper, 2026.

Conference Papers

- **Martirosyan, V.** et al. “Testing of Linux File System Drivers.” *Proceedings of ISP RAS*, vol. 23, Moscow, Dec 2012.
- **Martirosyan, V.** et al. “The Spruce System: Quality Verification of Linux File System Drivers.” *SYRCoSE 2012*, May 2012.
- **Martirosyan, V.** et al. “Dynamic Verification of Linux File System Drivers.” *CSIT 2011*, Sep 2011.
- **Martirosyan, V.** et al. “The Spruce System: Linux Kernel Module Verification Automation” (in Russian). *7th Annual Conference of RAU*, Dec 2012.
- **Martirosyan, V.** et al. “Fault Simulation in Linux Kernel Module Verification” (in Russian). *7th Annual Conference of RAU*, Dec 2012.

- **Martirosyan, V.** et al. “Linux Verification – Challenge of Today.” *7th Annual Conference of RAU*, Dec 2012.
- **Martirosyan, V.** et al. “X2C – XML To Code: Advantages of Standards.” *4th Scientific Annual Conference of RAU*, Dec 2009.
- **Martirosyan, V.** et al. “X2C: Code Generation Technology.”

Books & Tutorials

- **Martirosyan, V.** *Programming via Operating Systems, Second Edition*. Hilfmann Press, Sep 2024.
- **Martirosyan, V.** et al. *Tutorial: Basics of Programming in OS Linux Environment* (in Russian). MIPT, Apr 2011.
- **Martirosyan, V.** et al. *Tutorial: Programming in GNU/Linux Environment – From System Calls to the GUI* (in Russian). RAU Publishing, Dec 2010.

Translations

- *Jinnetic Engineering* by Richard Stallman, translated into Armenian. Dec 2010.

Research & Industry Experience

Research

Chief Scientist

Jan 2026 – Present

ExplainableAI.app

- Directing scientific strategy and model architecture for a no-code ML platform built entirely on intrinsically interpretable polynomial regression models — no black-box neural networks.
- Platform combines browser-side WebAssembly training (data never leaves the device), a hand-written C++ inference engine with sub-millisecond latency, and an AI App Builder (powered by Gemini) that turns any trained model into a shareable live web app.
- Applicable to any domain: users train models on any CSV or image dataset and deploy them as shareable live apps — no code, no infrastructure required.

Chief Executive Officer

Jun 2022 – Jul 2023

System View Armenia, Yerevan

- Developed LLVM/Clang-based auto-vectoriser for Xilinx ACAP (Visual System Integrator toolchain).

Head of System Programming Laboratory

Jan 2009 – Jul 2013

Russian-Armenian University / ISP RAS, Yerevan

- Led RAS System Programming Lab; directed Linux Foundation verification of system libraries and kernel modules.

Junior Research Scientist

2003 – 2005

Institute of Mathematics, National Academy of Sciences of RA

- Parallelisation of numerical algorithms for integral equations using MPI.

Industry (Selected)

Director of Software Engineering

Sep – Dec 2023

Plat.AI, Yerevan

Associate Software Development Manager <i>HelpSystems Armenia, Yerevan</i>	Oct 2021 – Jun 2022
Principal Software Engineer / Team Lead <i>HelpSystems Armenia, Yerevan</i>	Feb 2017 – Oct 2021
Software Architect (Contract) <i>Priotix, Yerevan · smart-home IoT system</i>	2018 – 2020
Principal / Senior Software Developer <i>Optym, Yerevan · logistics optimisation</i>	Aug 2015 – Feb 2017
Senior Software Developer <i>ArdInnoTekh LLC, Yerevan</i>	Aug 2013 – Jul 2015
Software Developer <i>ISMoTech, Yerevan · firmware, C++/C#, ASP.NET</i>	Oct 2006 – Dec 2008
Software Developer <i>Lycos Europe · PHP/MySQL</i>	Mar 2005 – Oct 2006

Professional Development & Certifications

AI, Machine Learning & Data Science

- Fundamentals of Deep Learning *NVIDIA, Apr 2025*
- Artificial Intelligence Foundations: Thinking Machines *LinkedIn, Jul 2024*
- Introduction to Artificial Intelligence *LinkedIn, Jul 2024*
- Advanced NoSQL for Data Science *LinkedIn, Jan 2025*
- Statistics Foundations 1: The Basics *LinkedIn, Sep 2024*

Embedded Systems, FPGA & Hardware

- C Programming for Embedded Applications *LinkedIn, Nov 2024*
- Introduction to FreeRTOS and Basic Task Management *LinkedIn, Nov 2024*
- Getting Started with RISC-V *LinkedIn, Jul 2024*
- Learning Verilog for FPGA Development *LinkedIn, Jul 2024*
- Learning FPGA Development *LinkedIn, Jul 2024*
- Introduction to Quantum Computing *LinkedIn, Jul 2024*

Software Architecture & Engineering

- Software Architecture: Domain-Driven Design *LinkedIn, Jul 2024*
- Microservices Foundations *LinkedIn, Jul 2024*
- Microservices: Asynchronous Messaging *LinkedIn, Jul 2024*
- Microservices: Security *LinkedIn, Jul 2024*

Leadership & Communication

- How to Be an Effective Manager to Drive Impact *LinkedIn*, Jul 2024
- Storytelling to Influence Leadership and Decision Makers *LinkedIn*, May 2024
- Improving Your Thinking *LinkedIn*, Jul 2024
- Leading with Emotional Intelligence *LinkedIn*, Apr 2019
- Public Speaking Foundations *LinkedIn*, Feb 2019